

Data Structures LAB

LAB - 03

1. Implement a menu-driven program in C / C++ to perform the following operations on an array:

- (a) Insert(POS, EL) (b) Delete(POS)
- (c) Search(EL) (d) Traverse
- (e) Update1(POS, NEW_VAL) (f) Update2(OLD_VAL, NEW_VAL)

Take **size of array** and **elements of array** as inputs from user. Take **POS**, **EL**, **NEW_VAL**, and **OLD_VAL** as inputs from user while performing the specific operations.

2. Take a Polynomial of order N as input from user and represent it using arrays.

3. Implement a program in C / C++ to initialize a array of size **N**. Take **N** and **elements of array** as inputs from user. Assuming data type of elements as **double**, determine the address of the element at index **POS** in the array. Take **POS** as input from user and $0 \leq POS < N$.

4. Perform a Right Circular Rotation operation on an array of positive integers **A** [**A**₀, **A**₁, **A**₂, **A**₃,, **A**_{N-1}]. After performing one Right Circular Rotation operation, the array is transformed from **A** [**A**₀, **A**₁, **A**₂, **A**₃,, **A**_{N-1}] to **A** [**A**_{N-1}, **A**₀, **A**₁, **A**₂,, **A**_{N-2}]. Perform this operation on array of **N** elements **K** times. Now, to identify elements at particular positions in the rotated array, ask **Q** queries where each query consists of a single integer **M**, which represents a particular index for an element in the rotated array. Implement a program in C / C++ for solving this problem.

5. A matrix can be represented using a two dimensional array **MAT[M][N]**, where **M** and **N** specify the number of rows and columns respectively. Each element of **MAT** is represented as **EL[I][J]**, where $0 \leq I < M$ and $0 \leq J < N$. You have to compute the product of two elements of **MAT**: **EL1[I1][J1] X EL2[I2][J2]**. Implement a program to solve this problem. Take **M**, **N**, and **elements of matrix** as inputs from user. For multiplication, take the indices of both elements **I1**, **J1**, **I2**, and **J2** as inputs from user. Implement a program to solve this problem using row-major ordering and column-major ordering.

Example:

1	2	3	4
8	7	6	5
8	7	6	5
1	2	3	4

For above matrix, **M = N = 4**. Suppose you have to compute product of **3** and **7**; as highlighted in the matrix. So, **I1 = 0**, **J1 = 2**, **I2 = 2**, and **J2 = 1** in **row-major ordering**. The output of multiplication is **7 X 3 = 21**.